Semi-Supervised Citation Function Classification Using GAN-CITE

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Abstract

Citation function classification is main task in bibliometric analysis and scientific knowledge discovery, yet it is constrained by the scarcity of labeled citation data. This paper presents and evaluates **GAN-CITE**, a semi-supervised framework that combines SciBERT contextual embeddings with Generative Adversarial Network (GAN) training to classify citation functions under less supervision. We train and evaluate the framework on the AlgoCite dataset using a ten-fold stratified cross-validation protocol at 18 labeled-data budget levels (1%–90%). We introduce three targeted training improvements: (1) a linear warm-up and decay learning rate scheduler, (2) macro-F1-based model selection with class-weighted loss, and (3) early stopping on validation macro-F1. These modifications increases the accuracy to **92.46%**, demonstrating that principled training strategies give substantial gains over the base GAN framework even without architectural changes.

Keywords:
Citation function classification, Semi-supervised learning, GAN, NLP, SciBERT, Early stopping, Learning rate scheduling

1. Introduction

Scientific research grows through links between papers, and these links are created through citations. When authors cite earlier work, they do more than just giving credit. They use citations to explain background ideas, describe methods they follow, compare results, or highlight gaps in existing studies. In other words, every citation plays an important role in defining the story of a research paper. Identifying this role, known as the citation function, helps us understand how scientific knowledge is built and how researchers support their reasonings.

Although citation function classification has been studied for many years, it is still a challenging task. The main problem lies in the need for careful labeled citation text. Setting the correct function to a citation requires domain knowledge. This makes the labeling process slow, and hard to scale. As new citation categories and definitions continue to appear, the need for labeled examples increases, but the amount of labeled data remains limited. As a result, many existing models struggle to perform well, particularly when trained on small or imbalanced datasets.

To solve this problem, we are working with a framework that uses semi-supervised learning to combine both labeled and unlabeled citation data. We make use of Generative Adversarial Networks(GANs) together with a language model trained on scientific text. This combination allows the model to learn better representations of citation contexts and to form clearer decision boundaries.

The related papers used for this study are listed in [1].

2. Problem Statement

Traditional citation function classification methods depend on large labeled datasets. However, creating these datasets needs manual labeling, which makes the process costly and time-consuming. Because there is a limited amount of labeled citation data, current supervised models often struggle with poor performance and class imbalance. Therefore, this project focuses on the question - **How can we classify citation functions accurately when we have only a small amount of labeled data?**

[1] Related work: `https://drive.google.com/drive/folders/1f9OHgMlXc6csXIRusBSygaeR-ZzS2CTD?usp=drive_link`

3. Objective

The main objective of this project is to implement and analyze the GAN-CITE framework for classification of citation function under limited labeled data conditions.

- To understand citation function classification and its challenges.
- To study datasets and perform data cleaning and exploratory data analysis.
- To implement the GAN-CITE semi-supervised learning framework.
- To evaluate the performance of the model under limited labeled data.

4. Dataset

4.1. Dataset Description

The GAN-CITE framework is reviewed on Algorithmic Citation data set of citation functions. This dataset is divided into labeled and unlabeled data. Fig. 1 shows class distribution across this dataset. To improve these labeled datasets, unlabeled data were prepared by randomly selecting samples from their original sources. The data for algorithmic citations was collected from CiteseerX repository (Wu et al., 2018).

Dataset	Categories (proportion)	Source	Samples
Algorithm citation functions (UTILIZATION scheme)	UTILIZE (13%), NOT-UTILIZE (87%)	CiteSeerX	8796
Algorithm citation functions (USAGE scheme)	USE (9%), EXTEND (4%), MENTION (65%), NOTALGO (22%)	CiteSeerX	8796

Figure 1: Citation function class distribution of Algo-cite dataset

Each instance of labeled data set includes not only the citation text but also detailed linguistic cue annotations that highlight suggestive words related to citation intent. Along with primary labels such as utilization and usage categories, the dataset provides detailed cue features capturing verbs (e.g., “proposed”, “use”), subjects (e.g., “algorithm”, “model”), comparative terms (e.g., “contrast”), tense information, and document-level keywords. These annotations shows the linguistic patterns and serve as a base for training models to distinguish differences between citation types.

The unlabeled dataset consists of citation contexts extracted from large collections of scientific papers, where each instance includes the raw citation context text without any annotation about its citation function. These samples share the same structural format as the labeled data but lack utilization, usage, and linguistic cue labels. Despite the absence of clear supervision, these contexts contain linguistic patterns typical of citation behavior, making them highly suitable for semi-supervised learning through the GAN framework. Figure 2 shows the comparison between labeled data and unlabeled data

Labeled data	Unlabeled data
Citation context text	Citation context text
UTILIZE / NOTUTILIZE	UNK_UNK
Usage labels	None
20+ cue annotations	None
Small in size	Very large

Figure 2: Comparison between labeled and unlabeled data

The dataset used in this study [2]

4.2. Data Cleaning

Before model training, the citation datasets must be cleaned and preprocessed to ensure data quality.

The following steps are performed:

- Initial review revealed noisy textual features due to PDF extraction and several additional columns not required for transformer-based modeling. So, we removed <cite>, performed lower casing, removed weird symbols, fixed broken multiple spaces, strip leading or trailing spaces as shown in Fig. 3

[2]Dataset: `https://github.com/snehach0909/Gan-cite`

```
]: print("Before:\n", df_train_30["CITATIONS_CONTEXTS"].iloc[0])
   print("\nAfter:\n", df_train_30["clean__citation_context"].iloc[0])

   Before:
    minates, a new seed is placed within the bundle and a new ber is advected. As de ned in Equation 2, the advection method for the be
   rs is accomplished using a simple fourth-order Runge-Kutta algorithm <cite>. To maintain the coherence of the bers, they must be adv
   ected in lock-step. This is important because as bers die o , they have to be re-seeded according to the distribution of the other b
   ers in the

   After:
    minates, a new seed is placed within the bundle and a new ber is advected. as de ned in equation 2, the advection method for the be
   rs is accomplished using a simple fourth-order runge-kutta algorithm . to maintain the coherence of the bers, they must be advected
   in lock-step. this is important because as bers die o , they have to be re-seeded according to the distribution of the other bers in
   the

]: df_train_30.columns
```

Figure 3: Comparative analysis of citation contexts.

- Length analysis revealed important variation in citation context sizes, ranging from shorter to longer paragraphs, so to normalize before model input, we removed citation contexts less than length of 300 as shown in Fig. 4

```
[55]: import matplotlib.pyplot as plt
      plt.figure(figsize=(6, 4))
      plt.hist(df_train_30["text_length"], bins=50)
      plt.title("Citation Context Length Distribution")
      plt.xlabel("Length")
      plt.ylabel("Count")
      plt.show()
```

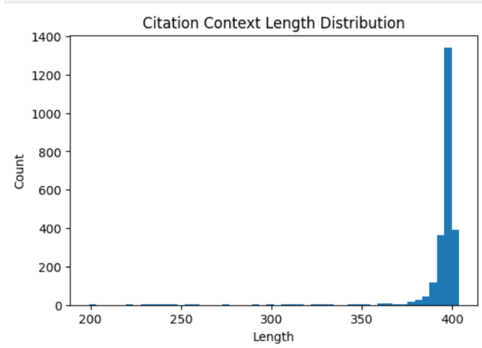

Figure 4: Histogram for length analysis of citation processed string.

- The task is a binary classification problem where citations are labeled as utilized or not utilized as shown in Fig. 5

```
[61]: df_train_30["label"] = df_train_30["UTILIZE_LABELS"].map({
          "UTILIZE": 1,
          "NOTUTILIZE": 0
      })
      df_train_30["label"].value_counts()

[61]: label
      0    1770
      1     630
      Name: count, dtype: int64
```

Figure 5: Showing value counts for each label.

- All cue-based features (CUE_WORDS, CUE_HUMAN, CUE_SUBJECT, etc.) are removed as shown in Fig. 6. We focus solely on the raw textual citation contexts and corresponding labels.(contextID, citationID, CITATIONS_CONTEXTS, UTILIZE_LABELS, and USAGE_LABELS)

```
[73]: # dropping all other columns
      df_train_30 = df_train_30[[
          "contextID",
          "citationID",
          "clean__citation_context",
          "utilize_label",
          "USAGE_LABELS"
      ]].copy()
      df_train_30
```

[73]:

	contextID	citationID	clean__citation_context	utilize_label	USAGE_LABELS
0	1809173	1915365	minates, a new seed is placed within the bundl...	0	MENTION
1	29908809	30604721	e an important category if we would like to ex...	1	USE
2	30522897	31235696	vitable for tasks whose domains are not well u...	0	MENTION
3	45673045	44790120	ound by random sampling. we can nd all regular...	0	MENTION
4	56602802	55932864	recursively: p i i 1hi (1.51)) 2 r1:i 1 pni p...	0	MENTION
...	...	...	...	...	...
2395	202587315	240514109	rastically from 12 to 24 h in the absence of l...	1	USE
2396	202277560	240071098	fers mainly due to its high cpu cost (figure 5...	1	USE
2397	202106589	239828417	al. 2002; adamic, lukose, and huberman 2003; w...	1	USE
2398	201946319	239593487	attenuation, the notations for p are similar, ...	1	USE
2399	201846185	239448101	out sampling from the uniform distribution san...	1	USE

2400 rows × 5 columns

Figure 6: CUE based features are dropped.

4.3. Data Preparation

Dataset preparation was done for two complete runs on different versions of the AlgoCite labeled dataset.

- **Initial Run:** conducted on a smaller 30% subset of AlgoCite labeled data (3,592 samples) with the main goal of validating the entire pipeline end-to-end. Results from this run were not reported in the paper.

- **Final Run:** conducted on the complete cleaned AlgoCite labeled dataset (8,931 samples) once the pipeline was fully correct and reliable. All experimental results reported in this paper are based on this run.

4.4. Overview of the Two-Stage Pipeline

The complete dataset preparation has two sequential stages applied identically to both runs.

4.4.1. Stage 1: Ten-Fold Stratified Cross-Validation

The first stage combines the three cleaned CSV files into a single unified labeled data pool. This dataset is then divided into ten stratified folds.

Stratified splitting is important due to class imbalance present in the AlgoCite dataset, where `NOTUTILIZE` citations outnumber `UTILIZE` citations. Without stratification, some folds may contain very few `UTILIZE` samples, leading to unstable training and inaccurate results.

For each of the ten folds, the dataset is divided as follows:

- **Test Split:** Approximately 10% of the entire dataset is split as test set.
- **Training and Validation Splits:** The remaining 90% of the data is further divided which results in approximately 72% of the full dataset used for training and 18% used for validation.
- **Output Files:** Each fold produces three CSV files named `algocite_utilize_dataset_{split}_fold_{n}.csv`. Each file contains the following columns: `contextID`, `citationID`, `clean_citation_context`, `UTILIZE_LABELS`, and `USAGE_LABELS`.

Fig. 7 shows per-fold data distribution where green means final dataset. The Class ratio is stratified and preserved across both runs.

Split	Proportion	Initial Run	Final Run	Class Ratio
Train	~72%	~2,585	~6,429	Preserved
Validation	~18%	~647	~1,608	Preserved
Test	~10%	~360	~894	Preserved

Figure 7: Per-fold data distribution

4.4.2. Stage 2: Percentage-Based Training Subsampling

The second stage focuses on an experimental question: how does model performance change as the amount of labeled training data is reduced? To explore different levels of supervision, the training portion of each fold is sampled at 18 different data budgets, ranging from (1%) to (90%).

The 18 budget levels are organized into two groups:

- **Fine-grained precent levels (1%–9%):** Nine densely sampled levels that capture the sharp part of the learning curve.
- **Coarse precent levels (10%–90%):** Nine more widely spaced levels that capture the flatter region of the learning curve.

Fig. 8 shows The 18 training precent levels used across both runs.

Budget Category	Percentage Levels
Fine-grained (1–9%)	1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%
Coarse (10–90%)	10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%

Figure 8: The eighteen training precent levels used over both runs.

4.5. Final Dataset Run

After the pipeline was validated, same procedure was applied to the complete cleaned AlgoCite dataset. Source files from `processed_data/final_dataset/` were combined into a pool of 8,931 samples which is approximately 2.49× larger than the initial dataset allowing reliable evaluation at all 18 levels of the training dataset. Fig. 9 shows the training sample counts per percent level for Fold 1 where green represents final dataset, grey represents initial dataset for comparison.

Budget	Final Train	UTILIZE	NOTUTILIZE	Val (full) | Test (full)	Initial Train
1%	63	15	48	1,608 | 894	~25
3%	190	37	153	1,608 | 894	~76
5%	318	75	243	1,608 | 894	~127
10%	636	~159	~477	1,608 | 894	~255
20%	1,272	~318	~954	1,608 | 894	~511
30%	1,909	~477	~1,432	1,608 | 894	~767
50%	3,182	~795	~2,387	1,608 | 894	~1,279
70%	4,454	~1,113	~3,341	1,608 | 894	~1,790
90%	5,780	~1,445	~4,335	1,608 | 894	~2,322

Figure 9: Training sample counts per precent level for Fold 1

4.6. Final Dataset Structure Summary

The 540 CSV files produced by Stage 2 are initially organized by budget level. However, the fine tuning notebook (`Finetuning_SciBert.ipynb`) expects the files to be arranged first by fold number and then by percentage. So, a spin-off folder was created by reorganizing the same files under a different folder. The same `StratifiedShuffleSplit` logic and random seed (42) are used in the spin-off files.

The final dataset preparation process gives the following structure, which is used for all experiments stated in this paper:

- **Base 10-fold folder:** 30 CSV files (10 folds $\times$ 3 splits) at full fold sizes, serving as the source for all Stage 2 subsampling.
- **Precent folders:** 18 folders of the form `10-fold_labeled_final/10-fold_labeled_increasing_relabel_ver3_{k}_percent`, where $k \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 40, 50, 60, 70, 80, 90\}$.
- **Files per precent:** Each percent folder contains 30 CSV files representing the *train*, *validation*, and *test* splits for each of the 10 folds.
- **Total files:** $18 \times 10 \times 3 = 540$ CSV files generated by Stage 2, in addition to the 30 base fold files.
- **Columns in each file:** `contextID`, `citationID`, `clean_citation_context`, `UTILIZE_LABELS`, and `USAGE_LABELS`.
- **Spin-off folder:** `spin-off/final_dataset/fold_{n}/10-fold_original_{k}_percent/`, including the same 540 files reorganized by fold number to match the requirements of the fine-tuning notebook.

The primary evaluation stated in this paper focuses on the 10%, 20%, and 30% percent levels, showing limited-data scenarios consistent with the original GAN-CITE framework. The full 1%–90% range is used for the learning curve analysis presented in the results section. All reported results including the SciBERT supervised baseline and GAN-CITE semi-supervised performance are calculated as 10-fold averages obtained from the 540 CSV files of the final dataset.

4.7. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the structure and characteristics of the dataset.

- There are two attributes in the dataset: a binary utilization label (UTILIZE_LABELS) and a finer-grained usage category (USAGE_LABELS). Analysis showed that the binary labels are derived from these detailed usage types, where “USE” and “EXTEND” correspond to utilized citations, and “MENTION” and “NOTALGO” correspond to non-utilized citations. The finer-grained labels were studied for exploratory analysis to understand citation behavior patterns as shown in Fig. 10.

```
[62]: pd.crosstab(df_train_30["UTILIZE_LABELS"], df_train_30["USAGE_LABELS"])
```

[62]:

USAGE_LABELS	EXTEND	MENTION	NOTALGO	USE
UTILIZE_LABELS				
NOTUTILIZE	0	1324	446	0
UTILIZE	157	0	0	473

Figure 10: Counts of UTILIZE_LABELS across USAGE_LABELS categories.

- Fig. 11 shows the data distribution for the "utilize_labels" in the dataset after mapped to 0 and 1 and Fig. 12 shows the data distribution for the "usage_labels" in the dataset for four different categories.

```
[71]: #data distribution for utilize_labels
import matplotlib.pyplot as plt
df_train_30['utilize_label'].value_counts().plot(kind='bar', color=['skyblue','salmon'])
plt.title('Distribution of UTILIZE vs NOTUTILIZE')
plt.xticks([0,1], ['NOTUTILIZE','UTILIZE'], rotation=0)
plt.ylabel('Count')
plt.show()
```

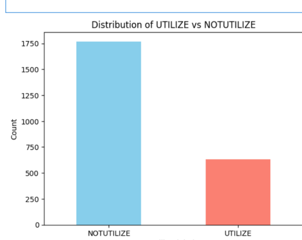

Figure 11: Data distribution of categories in utilize_labels

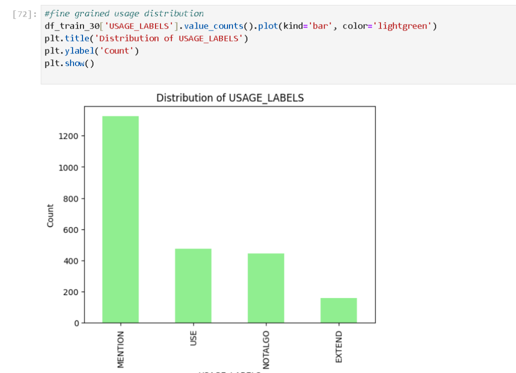

Figure 12: Data distribution of categories in USAGE_LABELS

The code for cleaning and analysis of the data [3]

5. Proposed Algorithm

This study suggests a **SciBERT + GAN based semi-supervised framework** for citation usage classification. The goal is to learn contextual representations of citation sentences on unlabeled data through adversarial training.

The architecture suggested includes of three main components:

- SciBERT Encoder
- Generator Network
- Discriminator Network

5.1. Citation Context Representation

Each citation context is changed into a textual sequence containing a special token <cite> that marks the citation position. Contexts are tokenized using the SciBERT tokenizer from the HuggingFace Transformers library. Each input gives three features: input IDs, attention masks, and token type IDs. The maximum sequence length is set to 256 tokens.

[3]Exploratory Data Analysis: `https://github.com/snehach0909/Gan-cite`

5.2. SciBERT Context Encoder

To capture semantic representations of text, the tokenized citation contexts are passed through the SciBERT transformer encoder. Since the custom token `<cite>` is added to the tokenizer vocabulary, the embedding layer is resized. SciBERT produces contextual embeddings with a hidden representation size of 768.

5.3. GAN-Based Semi-Supervised Learning

To make use of unlabeled data, the model integrates a Generative Adversarial Network (GAN) with two components.

5.3.1. Generator

The generator gives synthetic 768-dimensional embedding vectors that is similar to real SciBERT embeddings. It receives a random noise vector $\mathbf{z} \in \mathbb{R}^{100}$ as input and changes it through fully connected layers with LeakyReLU activation (slope 0.2) and dropout (rate 0.1).

5.3.2. Discriminator

The discriminator gets both real embeddings from SciBERT and fake embeddings from the generator. It simultaneously: (1) separates real embeddings from fake ones, and (2) predicts the citation usage class. The output logit layer has $K+1$ outputs, where K is the number of real classes and the extra dimension represents the "fake" class.

5.4. Training Objective

The discriminator is trained with three loss terms:

$$\mathcal{L}_D = \mathcal{L}_D^{sup} + \mathcal{L}_D^{unsup,real} + \mathcal{L}_D^{unsup,fake} \tag{1}$$

where $\mathcal{L}_D^{sup}$ is the supervised cross-entropy loss on labeled data, $\mathcal{L}_D^{unsup,real} = -\mathbb{E}[\log(1-p_{K+1}(\mathbf{x}))]$ penalizes real samples classified as fake, and $\mathcal{L}_D^{unsup,fake} = -\mathbb{E}[\log p_{K+1}(\tilde{\mathbf{x}})]$ penalizes fake samples not detected.

The generator is trained with:

$$\mathcal{L}_G = -\mathbb{E}[\log(1 - p_{K+1}(\tilde{\mathbf{x}}))] + \|\mathbb{E}_{\mathbf{x}}[\mathbf{f}(\mathbf{x})] - \mathbb{E}_{\tilde{\mathbf{x}}}[\mathbf{f}(\tilde{\mathbf{x}})]\|_2^2 \tag{2}$$

where the second term is a feature-matching regularizer that matches the mean features of real and fake embeddings.

6. Experimental Analysis

The experimental setup evaluated the impact of the proposed SciBERT–GAN framework for citation usage classification. It used SciBERT as the primary contextual encoder to represent citation contexts. When a citation context is given as input to the model, the SciBERT encoder processes the text and produces contextualized token representations. It generates a sequence of hidden vectors with a dimensionality of 768, which represent the contextual embedding of each token in the input sequence.

Combination of adversarial learning presents an additional form of regularization into the model training process. By forcing the discriminator to work in the presence of synthetic samples, the model learns to focus on more robust semantic features rather than relying on superficial patterns. This is beneficial for tasks involving complex linguistic structures, such as citation usage classification, where contextual differences figure out the role of a citation within a sentence.

The whole framework is implemented in PyTorch with the HuggingFace Transformers toolkit. Model parameters are optimized using AdamW, with a learning rate of 2×10^{-5} for both SciBERT and the GAN components. Training uses mini-batch gradient descent with a batch size of 32 and class-weighted cross-entropy loss to handle class imbalance. The GAN architecture uses one hidden layer each for the generator and discriminator, with a noise vector dimensionality of 100.

The primary evaluation focuses on the 10%, 20%, and 30% labeled-data percent levels as cases with limited labeled data. All reported results are 10-fold averages. The final predictions generated by the model are stored as structured output files contain the citation context, the predicted class label, and the related ground truth label.

7. Results

7.1. Finetuning SciBERT

This section presents the experimental results obtained from fine-tuning the SciBERT model for citation usage classification. The evaluation focuses on training convergence, classification performance, and class-wise prediction behavior.

The SciBERT model was fine-tuned for four epochs. Training loss decreased over time from approximately 1.12 in the first epoch to 0.76 by the fourth epoch, while validation loss decreased from 1.01 to 0.79. The small gap between training and validation loss shows that the model generalizes without meaningful overfitting.

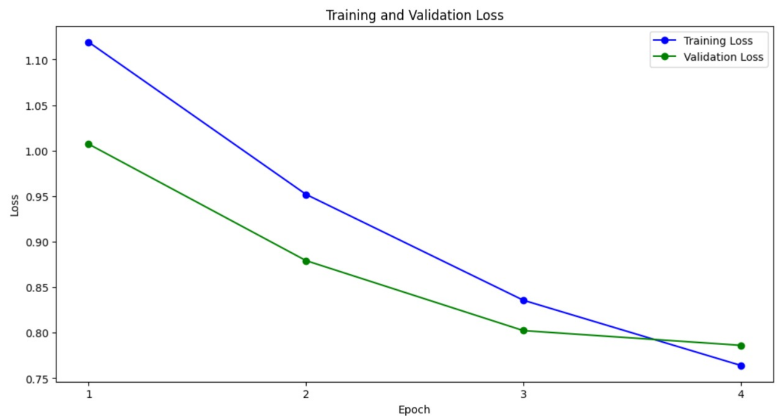

Figure 13: Training and validation loss curves during fine-tuning of SciBERT

The fine-tuned SciBERT achieved an overall classification accuracy of **69.49%**. Class-wise performance showed clear differences: the NOTALGO category performed the best, while smaller classes like USE and EXTEND were more difficult to tell apart. These results suggest that relying only on supervised fine-tuning with limited labeled data is not enough, highlighting the need for semi-supervised approaches.

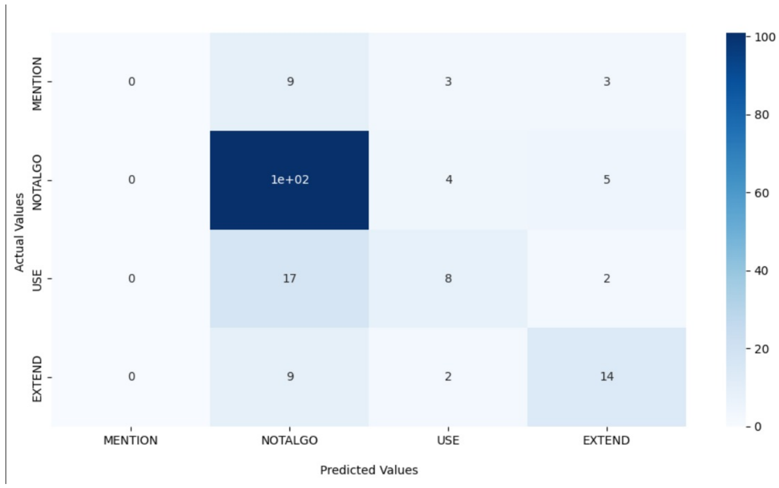

Figure 14: Classification performance of fine-tuned SciBERT

Overall, the results show that the fine-tuning of SciBERT provides a strong base for citation usage classification. However, performance gaps between classes indicate the need for improved representation learning and better handling of minority classes. So adversarial techniques such as GAN-based augmentation were used to increase the variety of training representations and improve the classification performance in experiments. In a second experiment with a larger training split, the loss values settled to lower levels (training: $1.04 \rightarrow 0.52$; validation: $0.83 \rightarrow 0.61$), and the NOTALGO class correctly classified approximately 220 instances while EXTEND got 72 correct predictions.

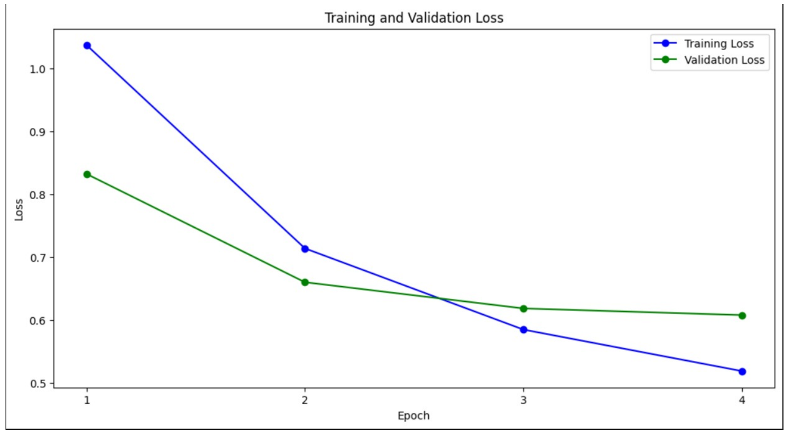

Figure 15: Training and validation loss during the second fine-tuning experiment

The decreasing trend in both training and validation loss shows that the model learns important representations of citation contexts during training. Compared with the first fine-tuning experiment, the loss values converge to lower levels, indicating that the model learns more smoothly and performs more consistently.

The confusion matrix for the second experiment is illustrated in Fig. 16. The matrix shows the distribution of the predicted classes against the true citation categories. The results explain a good classification accuracy for the *NOTALGO* and *EXTEND* categories. The *NOTALGO* class achieves the highest number of correct predictions, with approximately 220 correctly classified instances. Similarly, the *EXTEND* category shows strong performance with 72 correctly predicted samples.

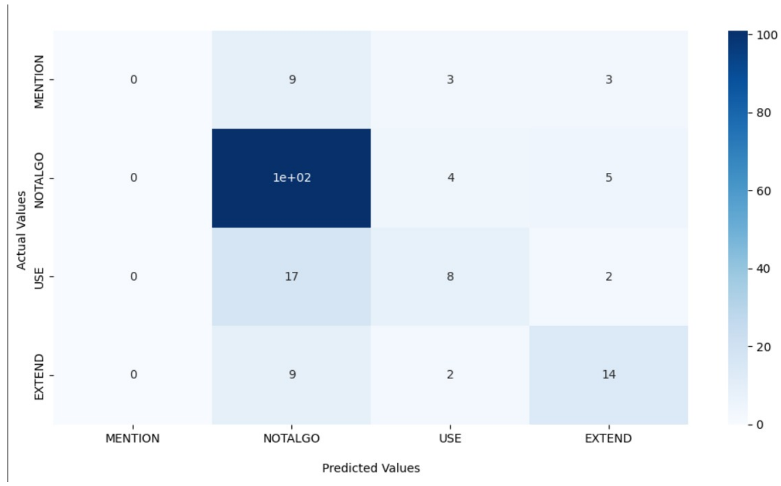

Figure 16: Confusion matrix for citation usage classification in the second fine-tuning experiment

Overall, the results of the second experiment show an improved classification behavior compared to the initial baseline. The confusion matrix indicates that the model is effective in identifying main citation categories, while minority classes are harder to identify correctly.

7.2. GAN-CFA Experiment

A Generative Adversarial Network with Citation Function Augmentation (GAN-CFA) was applied with the SciBERT embeddings. The generator and discriminator losses remained stable in 10 epochs (both fluctuating near 0.70), showing balanced adversarial training. Fig. 17 illustrates the generator loss, discriminator loss, and validation loss during training.

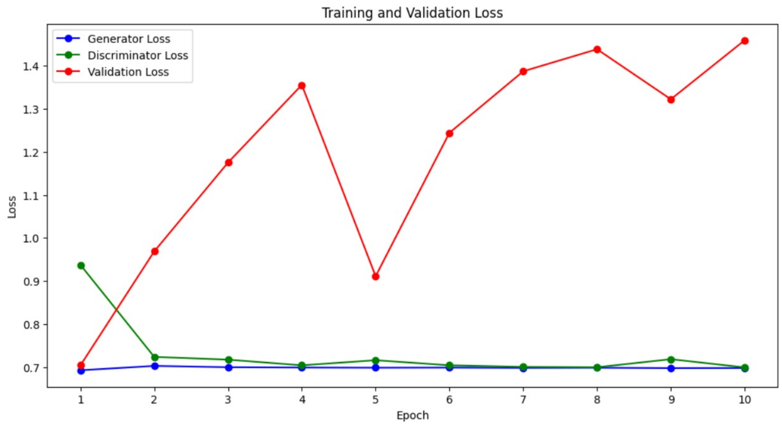

Figure 17: Generator loss, discriminator loss, and validation loss during GAN-CFA training

The GAN-CFA model achieved an overall performance of 85.23%, which shows improvement compared to the supervised baseline. The NOTUTILIZE class achieved the highest performance, while UTILIZE, the minority class, was classified with decent but lower accuracy, motivating further improvements to address class imbalance.

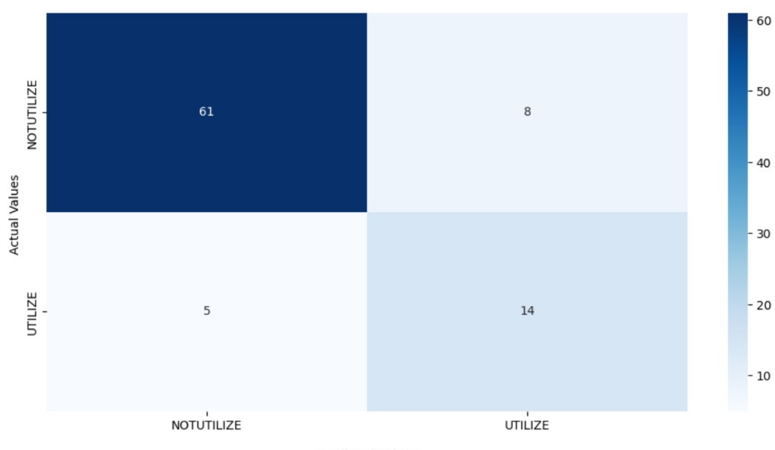

Figure 18: Confusion matrix for the GAN-CFA citation classification model

Overall, the GAN-CFA model shows improvements compared to the baseline fine-tuned SciBERT models. The adversarial training process improves the diversity of learned representations and allows the classifier to better understand the meaning of the citation. These results show that applying adversarial learning improves the way citations are classified, especially when we have limited labeled data.

All notebooks for SciBERT models and the GAN-CFA model are attached[4]

7.3. Improved GAN-CITE: Training Enhancements

To further improve performance, we introduced three training modifications over the GAN-CFA model:

7.3.1. Linear Warm-Up and Decay Scheduler

The original GAN-CITE uses a constant learning rate (with optional warm-up). We replaced this with `get _linear _schedule _with _warmup`, which linearly increased the learning rate from 0 to 2×10^{-5} over the first 10% of the training steps, then linearly decreased it to 0 at the end of the training. This is used for two purposes: the warm-up allows the GAN components to reach balance before full SciBERT updates begin, and the decay in later epochs prevents overfitting on the small labeled set. The scheduler is stepped once per batch for gradient update, not per epoch, which is the correct for transformer warm-up schedules.

7.3.2. Macro-F1 Model Selection with Class-Weighted Loss

The original model selects the best checkpoint based on validation performance. However, accuracy is misleading under class imbalance: a model that always predicts NOTUTILIZE achieves 87% accuracy while correctly classifying no UTILIZE instances. We replaced accuracy-based selection with macro-averaged F1 score, which calculates F1 per class and averages them,

[4]Google Colab links for models:
SciBERT fine-tune 1:
`https://colab.research.google.com/drive/1GOyO8TdXKN5oQmIgsGvy4DHFcUGl_GC0?usp=sharing`
SciBERT fine-tune 2:
`https://colab.research.google.com/drive/1skM__VcKVyYcCGPf95oEn7FTJ5BdbkNx?usp=sharing`
GAN-CFA:
`https://colab.research.google.com/drive/1Em29iPF2BzHC1Ck8FPwb3gySB2Z9Rkwl?usp=sharing`

penalizing models that ignore minority classes. Class-weighted cross-entropy loss is also used to handle imbalance during training.

7.3.3. Early Stopping on Validation Macro-F1

To avoid overfitting in longer training runs, we used early stopping with patience of 5 epochs. If the validation macro-F1 does not improve for 5 consecutive epochs, training stop and the best checkpoint is restored. This saves computation and avoids performance drop in later epochs.

7.3.4. Results of the Improved Model

Table 1 shows the per-epoch training statistics for the improved model trained on 90% labeled data (Fold 1).

Table 1: Training statistics for the improved GAN-CITE model (Fold 1, 90% labeled budget).

Epoch	**Gen. Loss**	**Disc. Loss**	**Val. Loss**	**Accuracy**
1	0.606	1.448	0.315	0.8966
2	0.708	0.850	0.286	0.9119
3	0.703	0.771	0.455	0.8882
4	0.702	0.735	0.417	0.9147
5	0.701	0.722	0.523	0.9113
6	0.700	0.713	0.666	0.8966
7	0.700	0.706	0.553	0.9210
8	0.700	0.705	0.679	0.9106
9	0.699	0.704	0.563	0.9231
10	0.700	0.703	0.626	0.9161

The improved model was evaluated on the unused test set (796 instances, Fold 1, 90% budget). The final test results are shown in Fig. 19.

```
Accuracy:  92.46231%
Precision: 90.79731%
Recall:    89.18906%
F1 Score:  89.94975%

              precision    recall  f1-score   support

  NOTUTILIZE      0.940     0.959     0.950       591
     UTILIZE      0.876     0.824     0.849       205

    accuracy                          0.925       796
   macro avg      0.908     0.892     0.899       796
weighted avg      0.924     0.925     0.924       796
```

Figure 19: Final test set performance of the improved GAN-CITE model.

Fig. 20 shows the confusion matrix with 567 true NOTUTILIZE correctly classified (24 misclassified as UTILIZE) and 169 true UTILIZE correctly classified (36 misclassified as NOTUTILIZE).

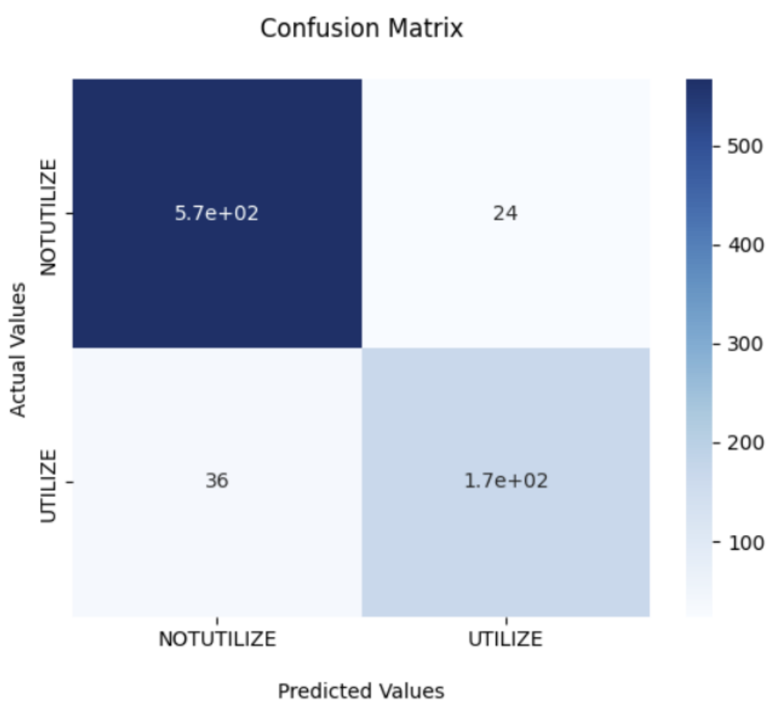

Figure 20: Per-class classification report for the improved GAN-CITE model.

The improved model achieves a **7.23 percentage-point** accuracy gain over the GAN-CFA original model and a **22.97 percentage-point** increase over the supervised SciBERT. This shows that training design, including proper learning rate scheduling, handling minority class, model selection, and early stopping, helps to achieve strong performance gains without changing the core model architecture. The code for improved model [5]

8. Conclusion

This paper presents and evaluates GAN-CITE, a semi-supervised citation function classification framework combining SciBERT and GAN-based adversarial training. It was tested on the AlgoCite dataset using ten-fold balanced cross-validation, the results show a steady improvement: a supervised SciBERT model reaches 69.49%, the GAN-CFA model increases it to 85.23%, and three principled training improvements further raise accuracy to 92.46% with a macro-F1 score of 89.95%.

The improvements—linear warm-up and decay scheduling, model selection based on macro-F1 with class-weighted loss, and early stopping—are easy to apply and highly effective. This suggests that they can be used as standard methods for training GAN-based semi-supervised text classifiers, especially when labeled data is limited.

Future work includes extending this approach to multi-class citation categories (e.g., the CL-SciSumm and SciCite schemes), exploring class-conditional generators to improve minority-class recall, and applying the method to larger scientific datasets beyond algorithm-related citations.

[5] Improved GAN-CITE: `https://colab.research.google.com/drive/1xujXvtg2qYZs1TQoUH-U7qUQICSyVOnu#scrollTo=32cAqLm9L9zN`

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